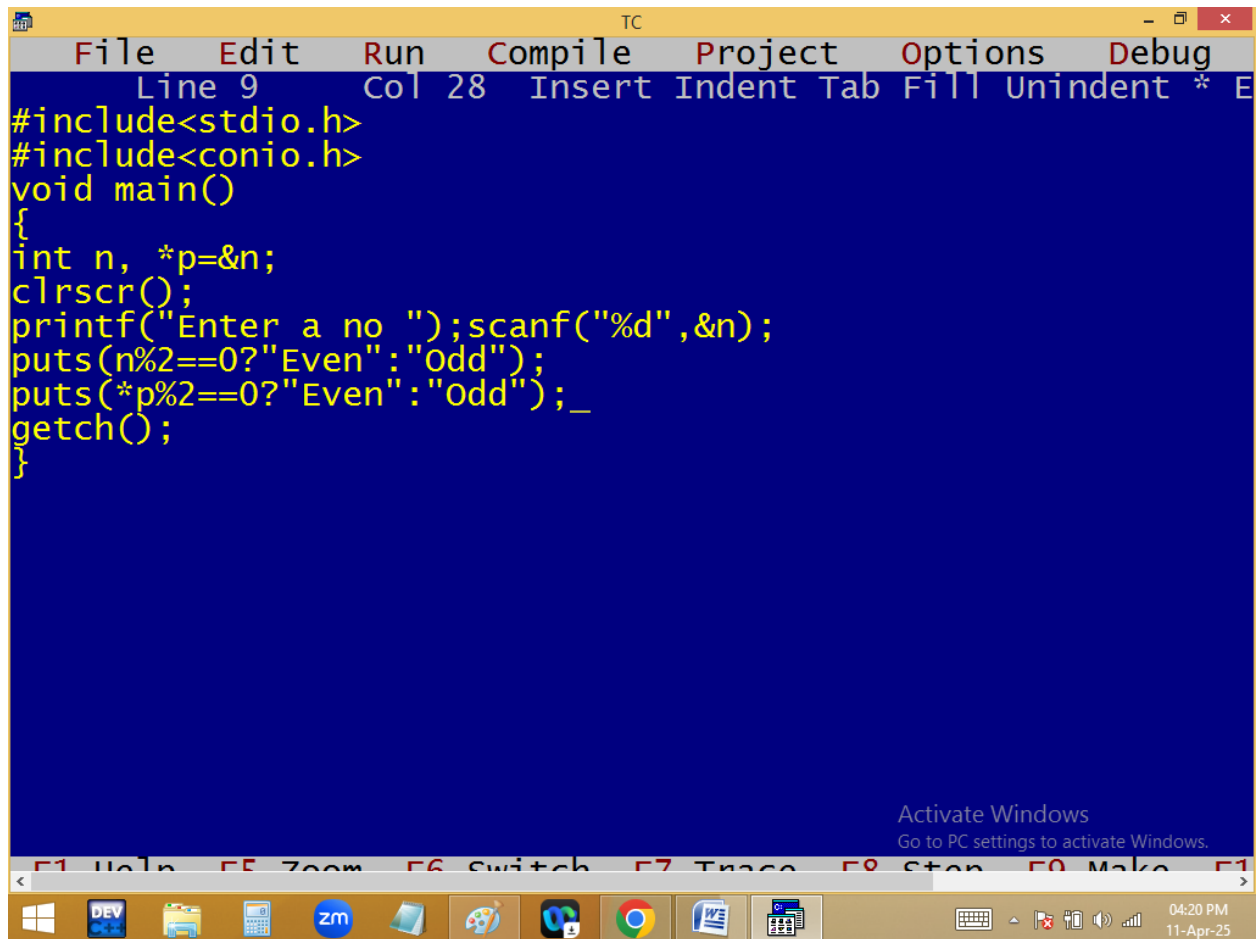


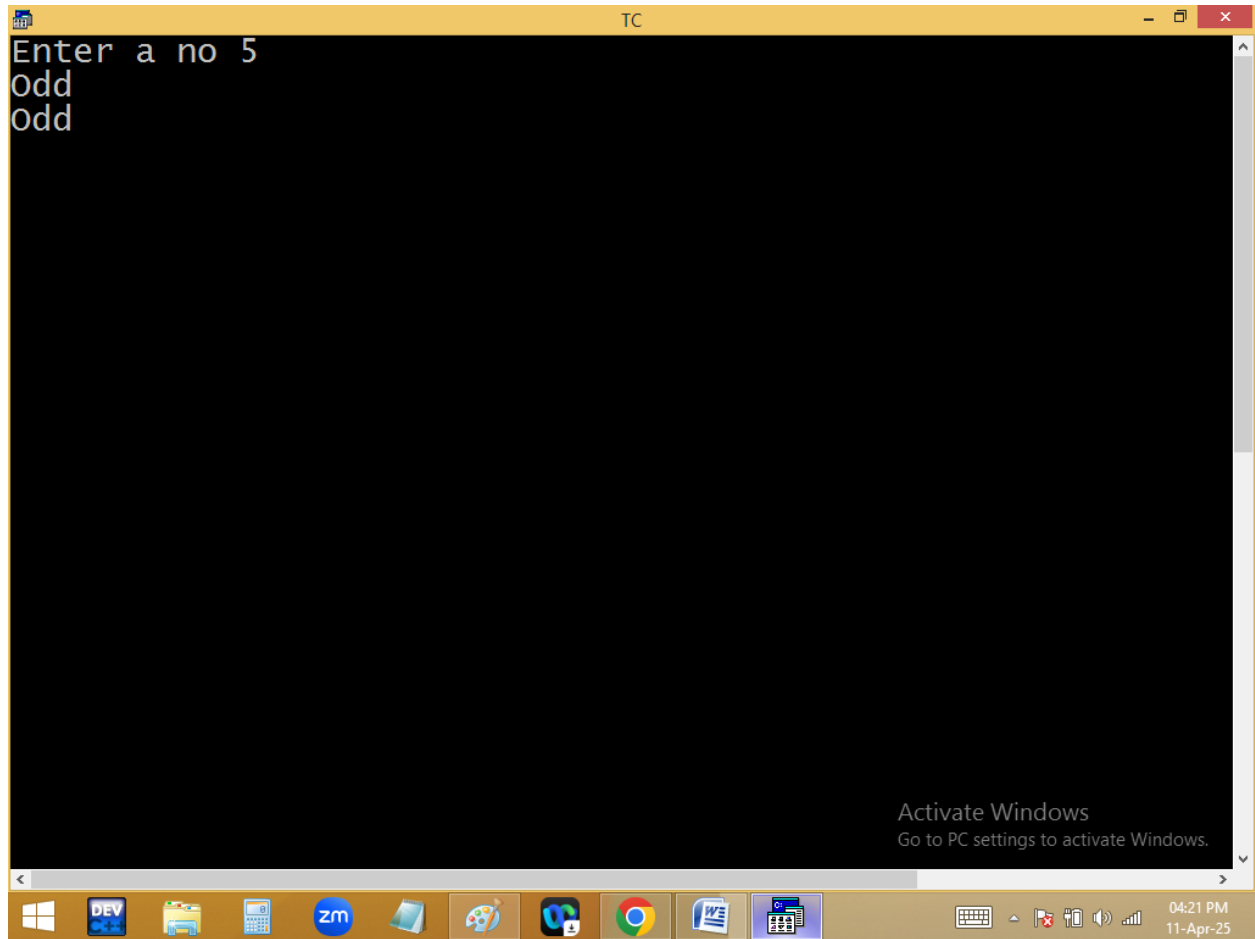
Finding even/odd using a pointer variable?



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 9", "Col 28", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n, *p=&n;
clrscr();
printf("Enter a no ");scanf("%d",&n);
puts(n%2==0?"Even":"Odd");
puts(*p%2==0?"Even":"Odd");_
getch();
}
```

At the bottom of the window, there is a taskbar with various application icons including Windows, DEV, File Explorer, Calculator, Zoom, and others. The system tray on the right shows the time as "04:20 PM" and the date as "11-Apr-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.



```
TC
Enter a no 8
Even
Even

Activate Windows
Go to PC settings to activate Windows.
```

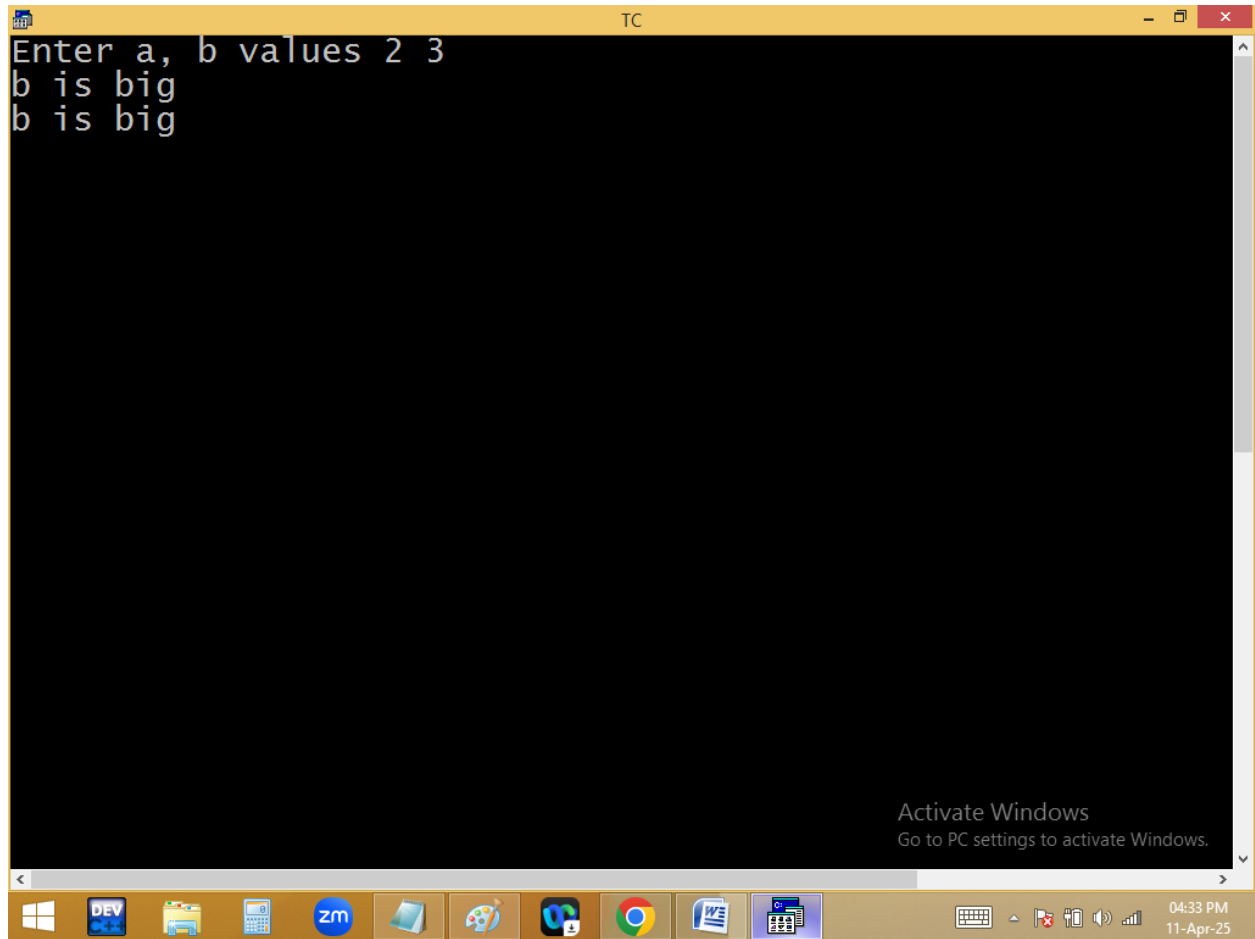
Finding max in 2 no's using 2 pointers?

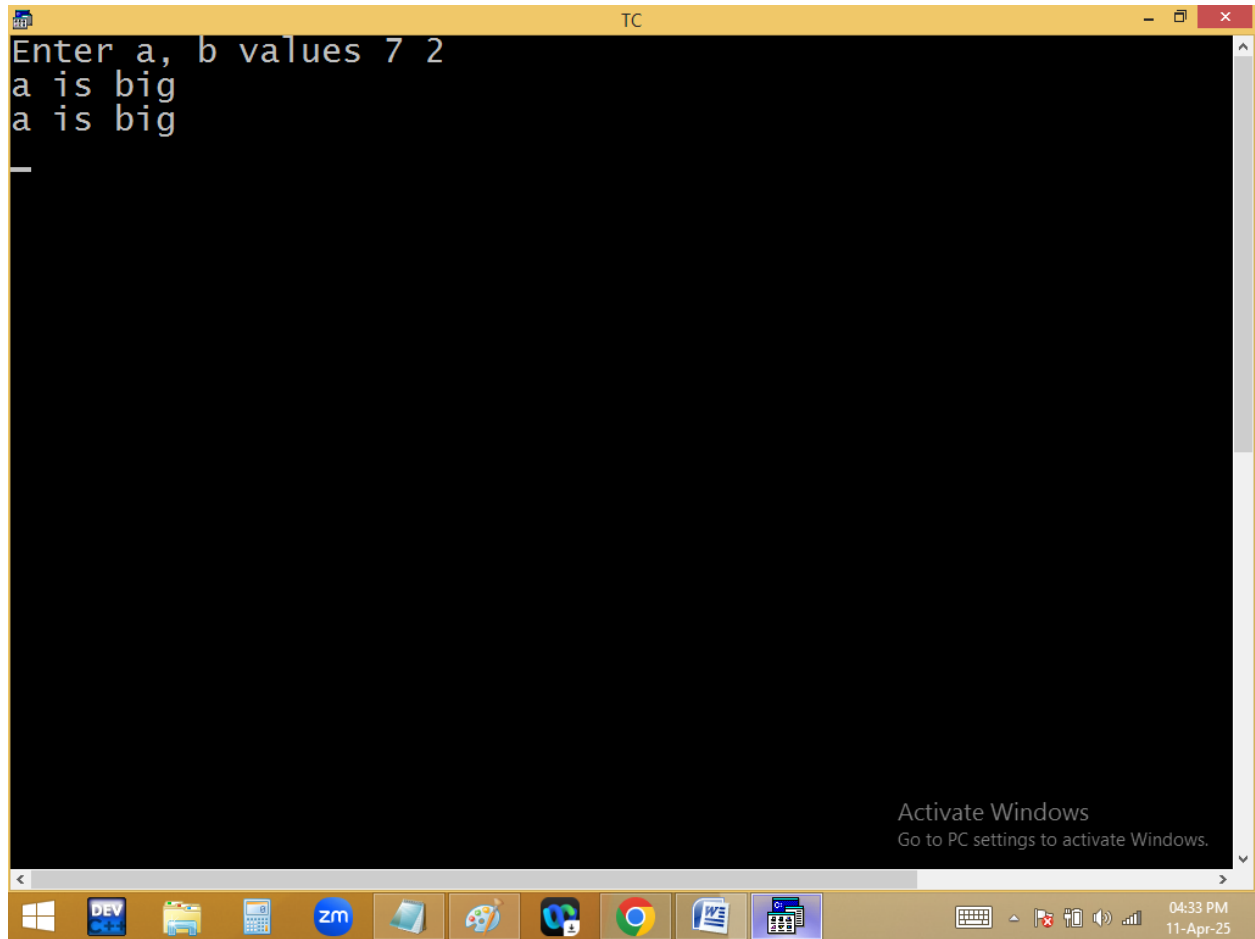
```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 28 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b, *p=&a, *q=&b;
clrscr();
printf("Enter a, b values ");scanf("%d%d",&a, &b);
puts(a>b?"a is big":b>a?"b is big":"both are Equal");
puts(*p>*q?"a is big":*q>*p?"b is big":"both are Equal");
getch();
}
```

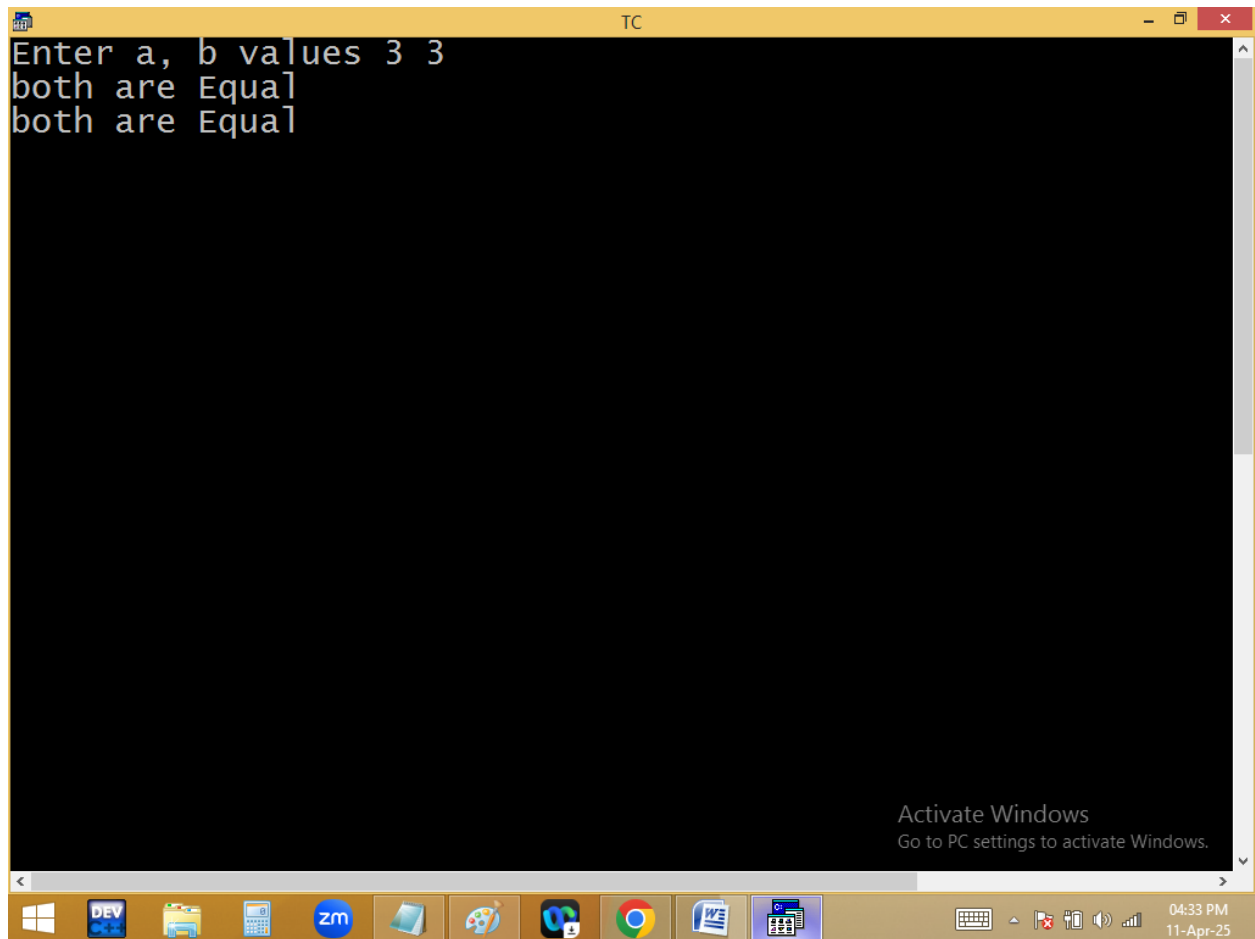
Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

04:33 PM
11-Apr-25



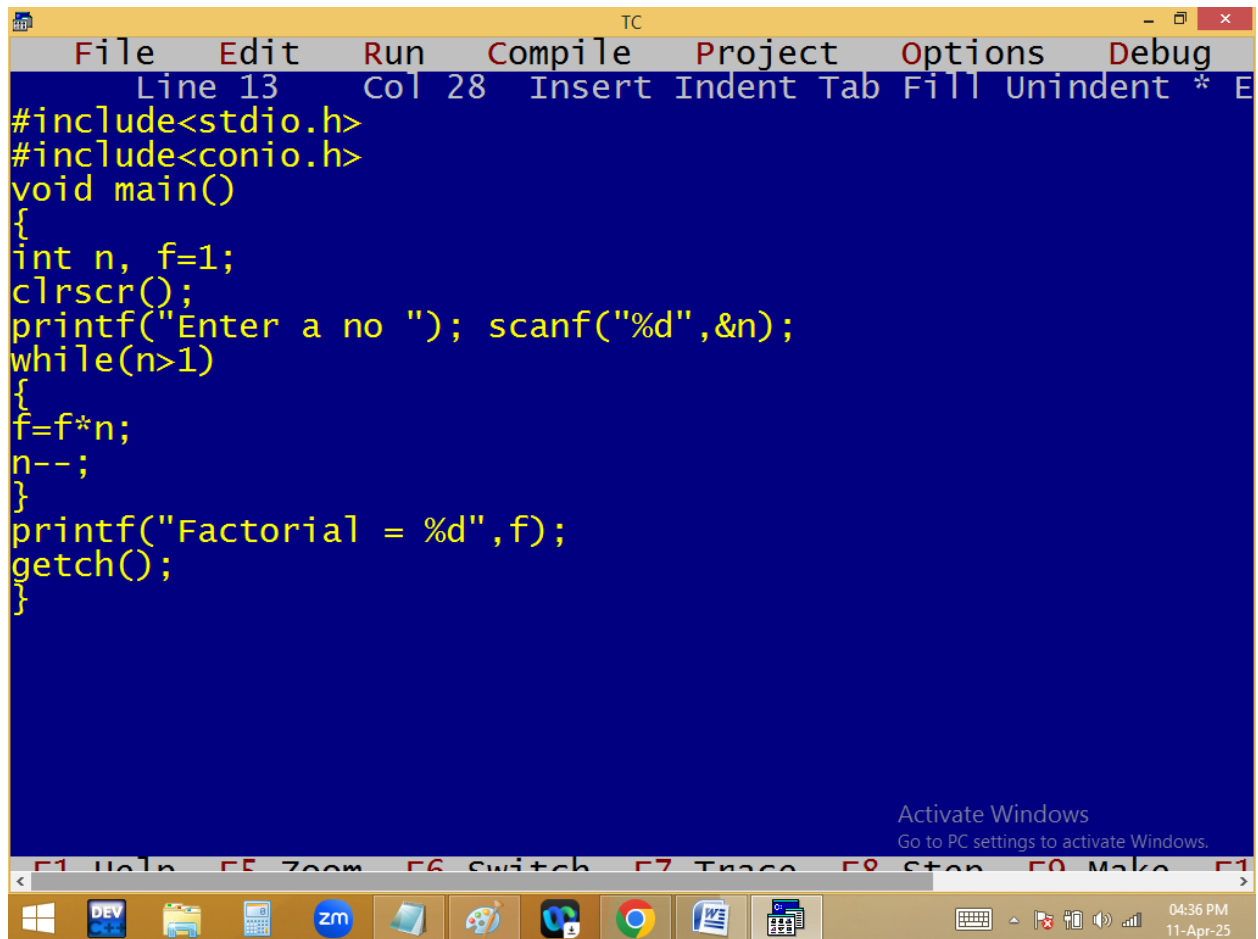




```
Enter a, b values 3 3
both are Equal
both are Equal
```

Activate Windows
Go to PC settings to activate Windows.

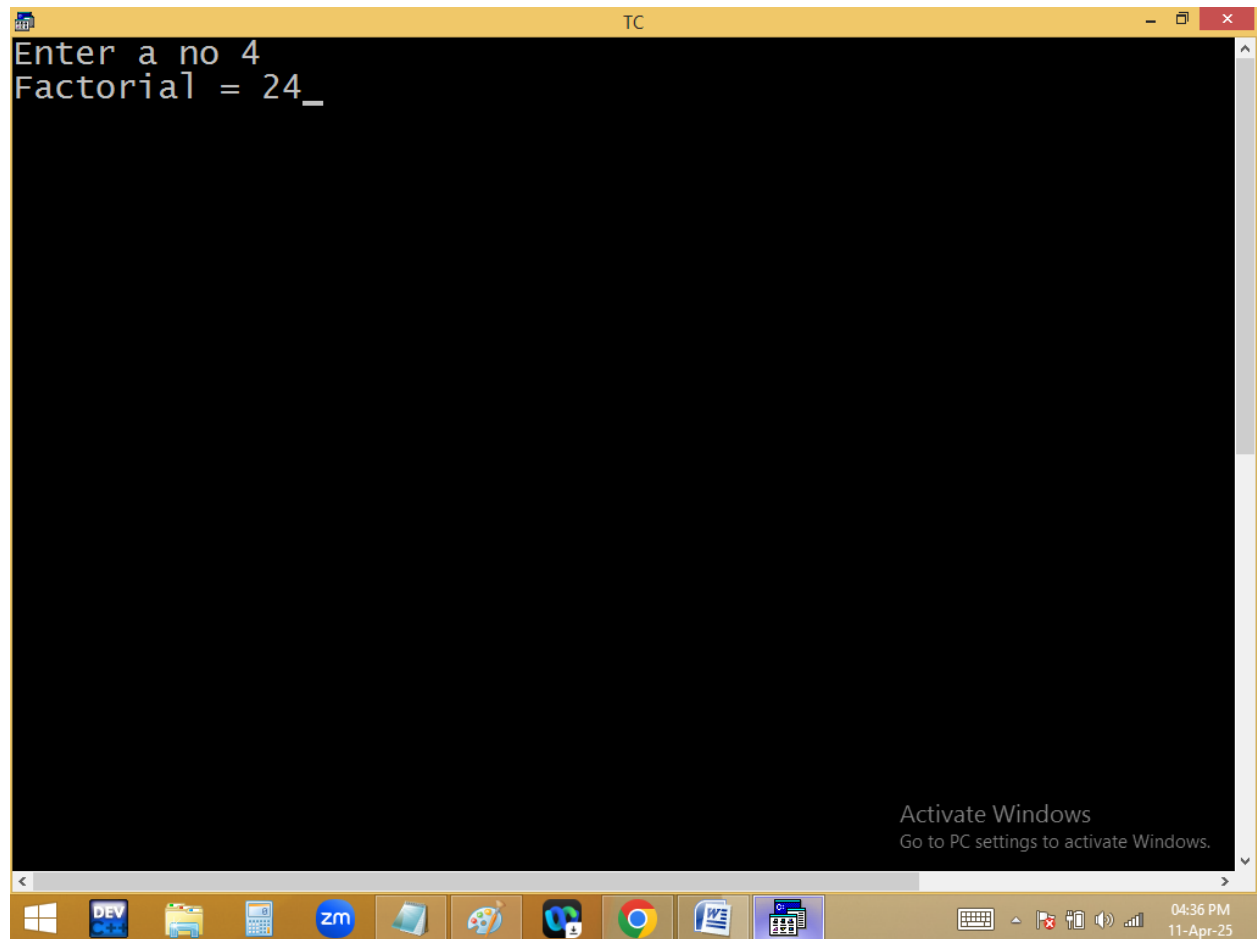
Finding factorial value using a pointer?

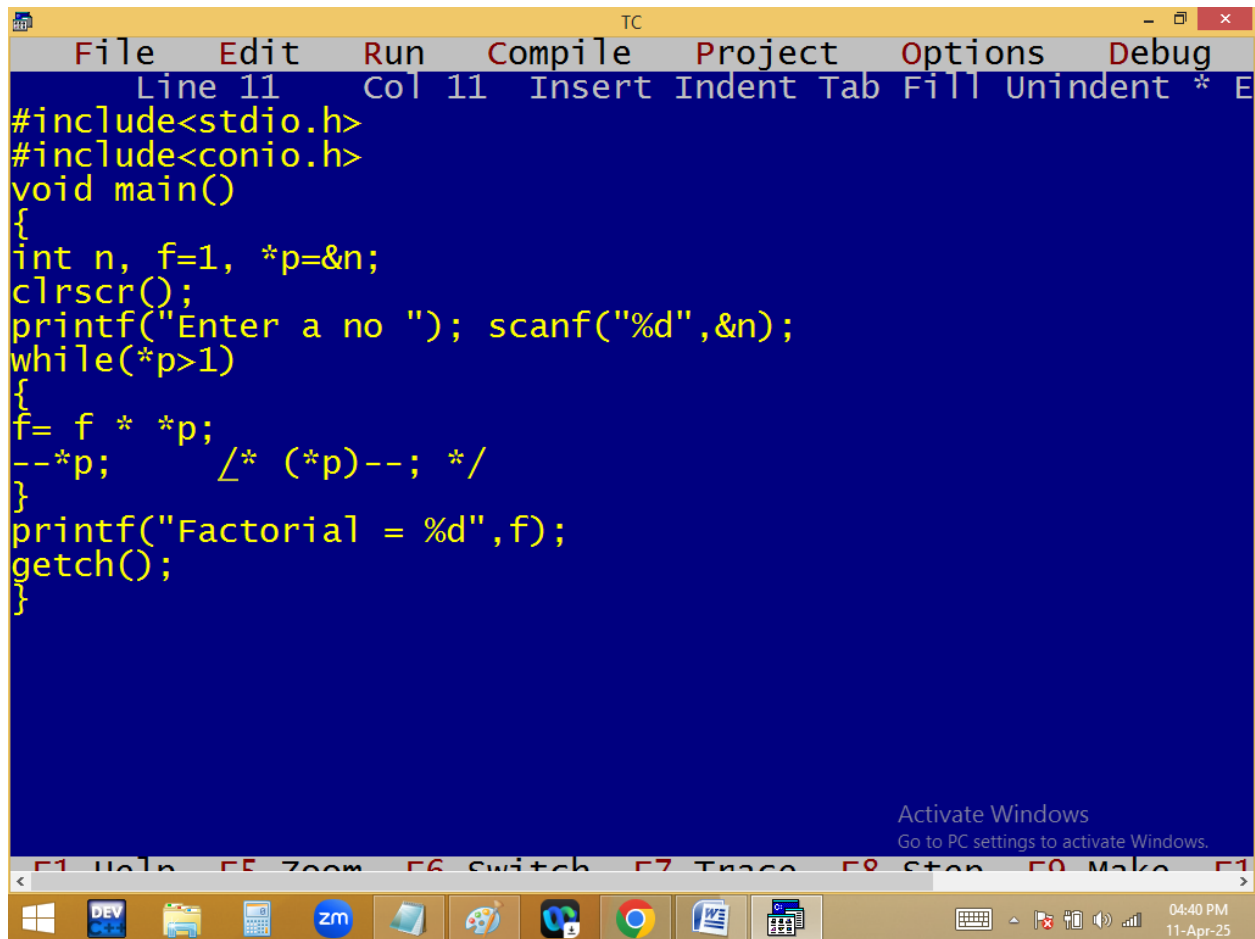


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 13", "Col 28", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n, f=1;
clrscr();
printf("Enter a no "); scanf("%d",&n);
while(n>1)
{
f=f*n;
n--;
}
printf("Factorial = %d",f);
getch();
}
```

At the bottom of the window, there is a taskbar with various application icons including Windows, DEV, File Explorer, Calculator, Zoom, and others. A system tray on the right shows the time as 04:36 PM and the date as 11-Apr-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

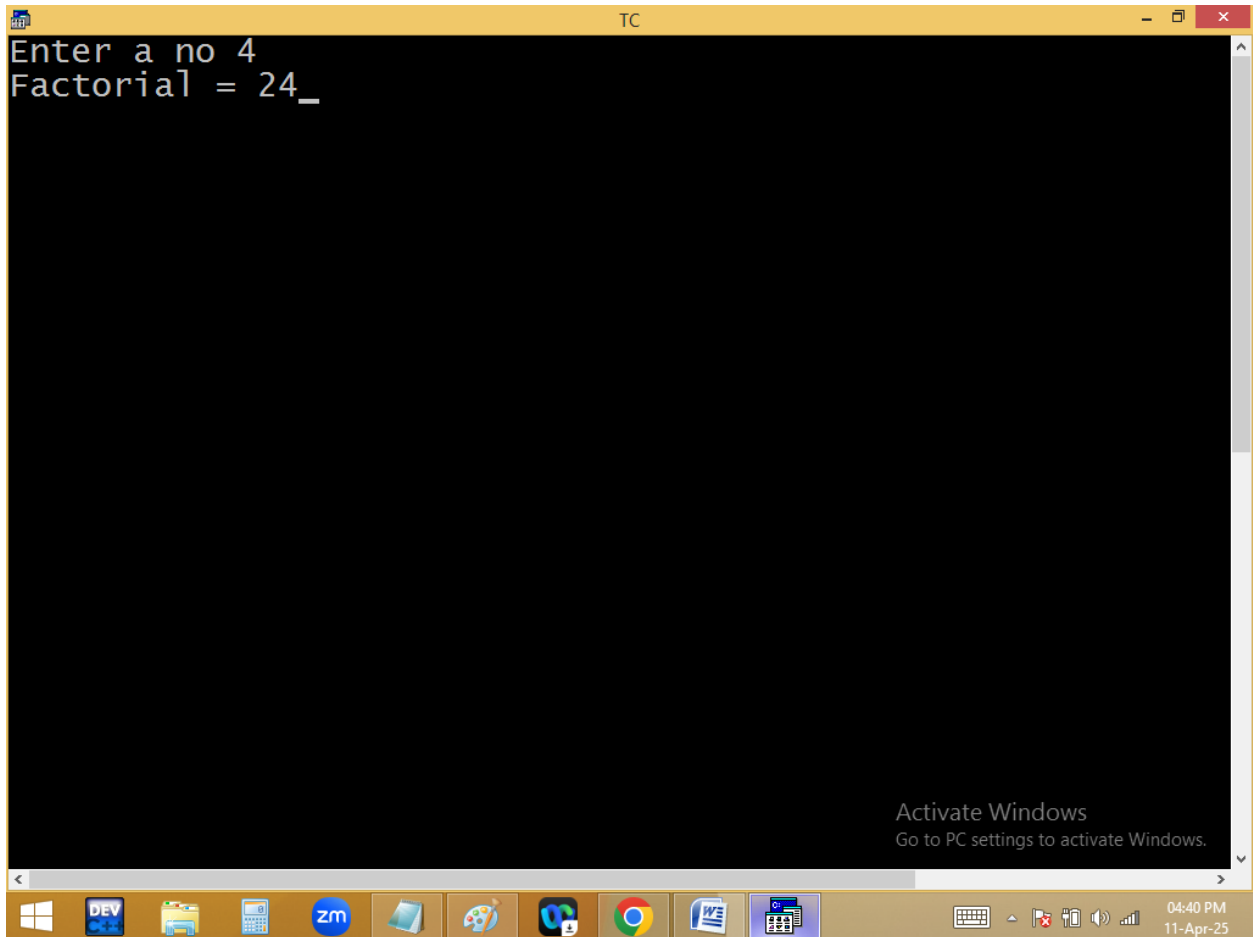




The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 11", "Col 11", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code for calculating a factorial:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n, f=1, *p=&n;
clrscr();
printf("Enter a no "); scanf("%d",&n);
while(*p>1)
{
f= f * *p;
--*p; /* (*p)--; */
}
printf("Factorial = %d",f);
getch();
}
```

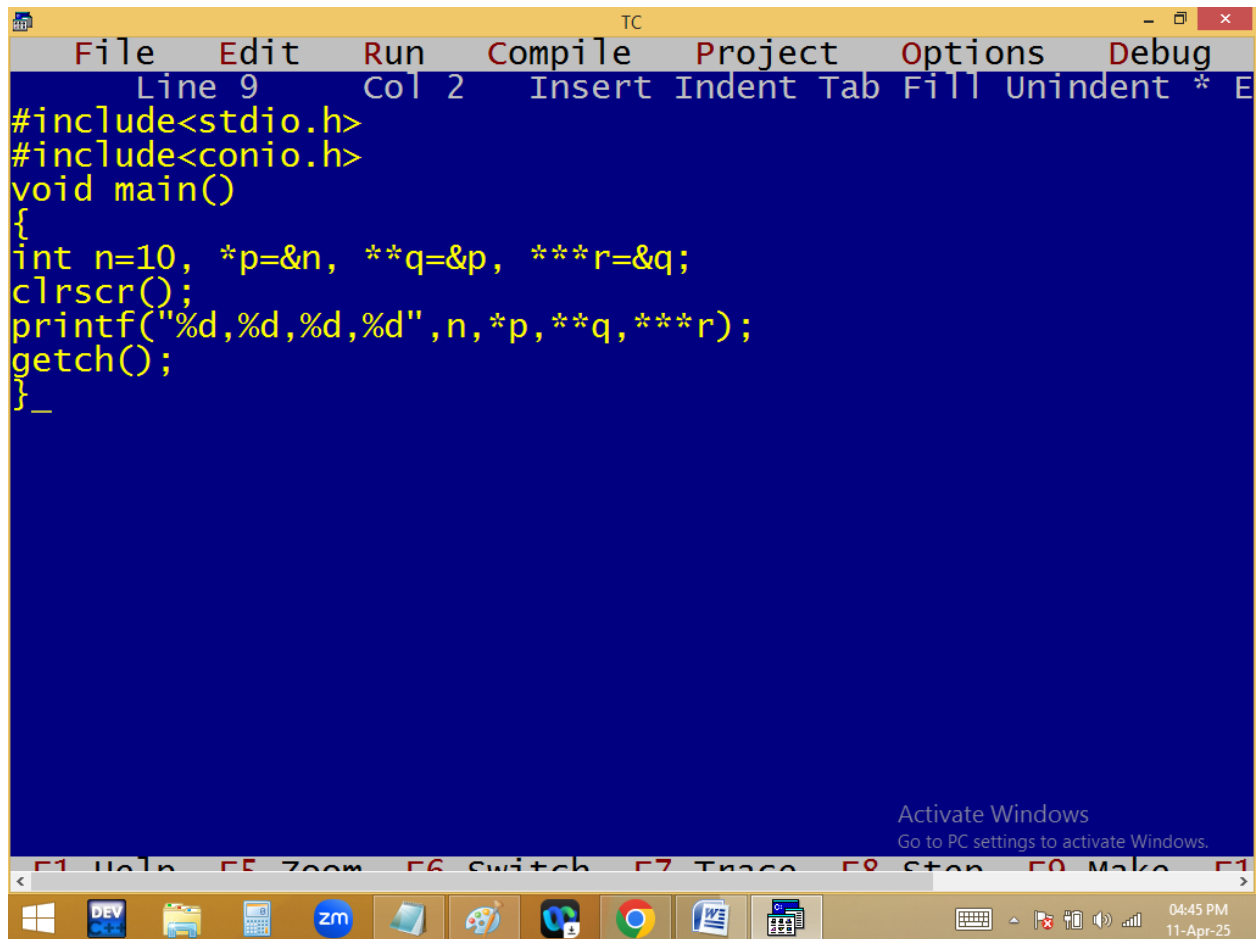
At the bottom of the window, there is a taskbar with icons for Windows, DEV, File Explorer, Calculator, Zoom, and several other applications. A system tray on the right shows the time as 04:40 PM and the date as 11-Apr-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.



The image shows a screenshot of a Turbo C++ (TC) compiler window. The window has a yellow title bar with the text "TC" and standard Windows window controls (minimize, maximize, close). The main area is black and contains the following text in white: "Enter a no 4" followed by "Factorial = 24_". The Windows taskbar is visible at the bottom, showing various application icons including DEV, File Explorer, Calculator, Zoom, and several others. The system tray on the right shows the time as 04:40 PM and the date as 11-Apr-25. An "Activate Windows" watermark is visible in the bottom right corner of the black area.

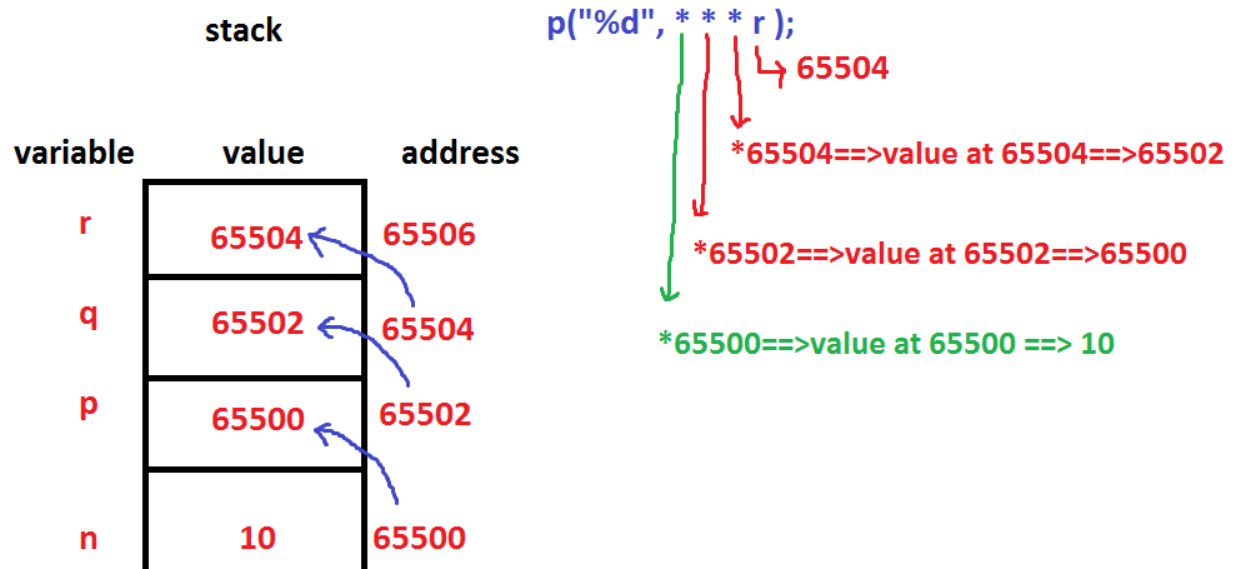
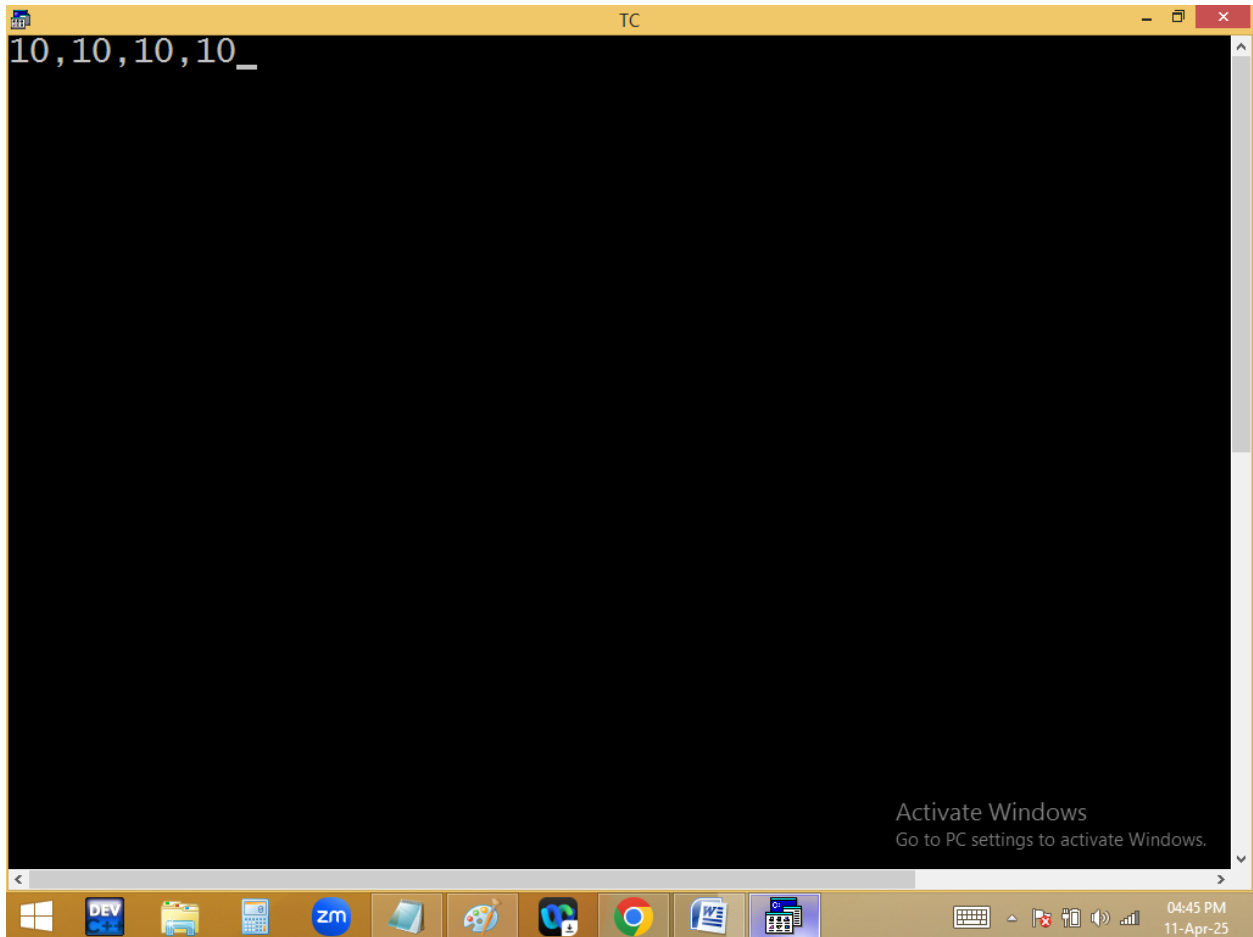
```
Enter a no 4
Factorial = 24_
```

Double pointer or pointer to pointer? The pointer which stores the address of another pointer is called double pointer or pointer to pointer. It is used to handle dynamic multi dimensional arrays.

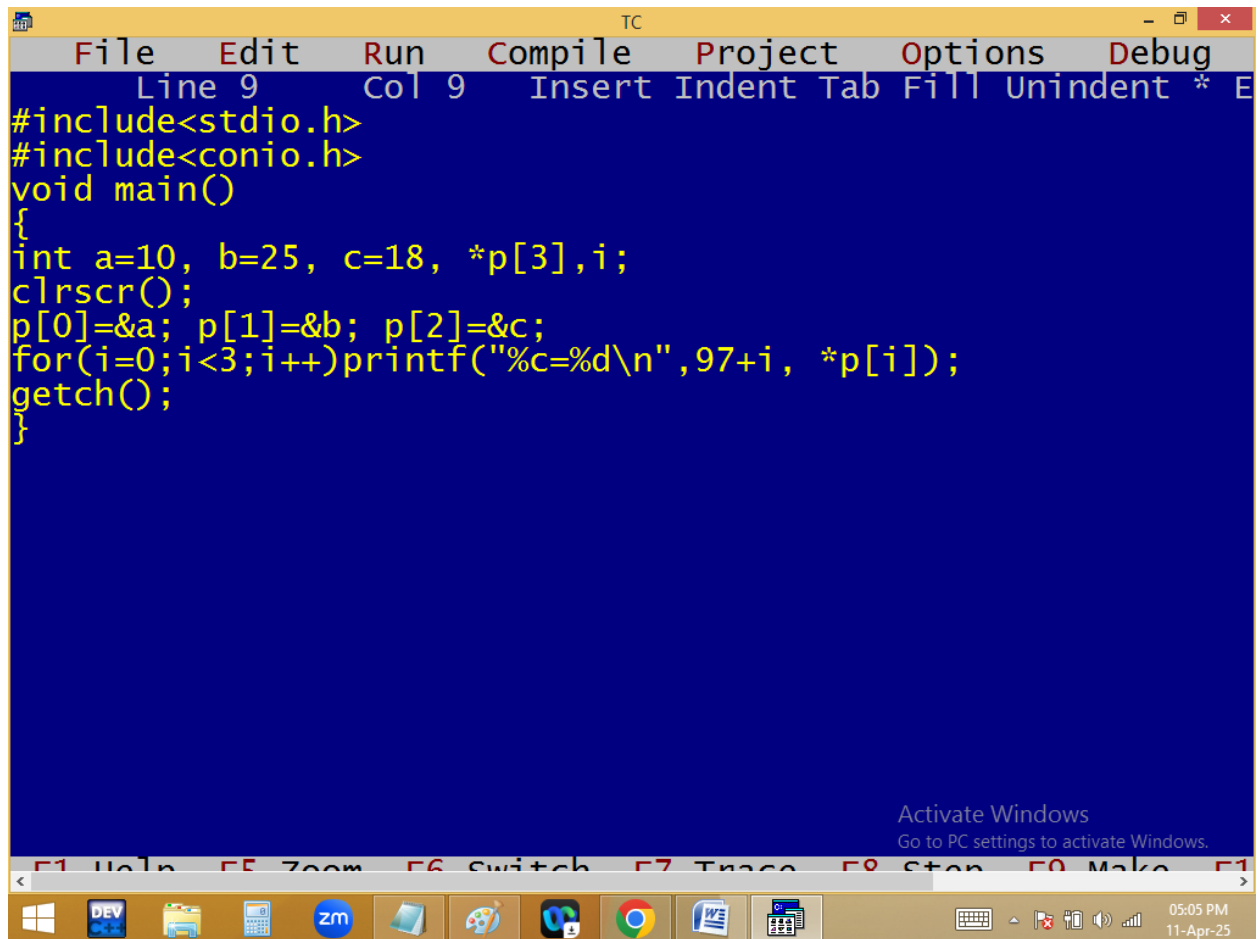


```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 2 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int n=10, *p=&n, **q=&p, ***r=&q;
clrscr();
printf("%d,%d,%d,%d",n,*p,**q,***r);
getch();
}_

Activate Windows
Go to PC settings to activate Windows.
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```



Array of Pointer: Like normal variables we can also declare pointer using array. Due to this the same pointer can store multiple variables address. It is used to handle dynamic multi dimensional arrays.



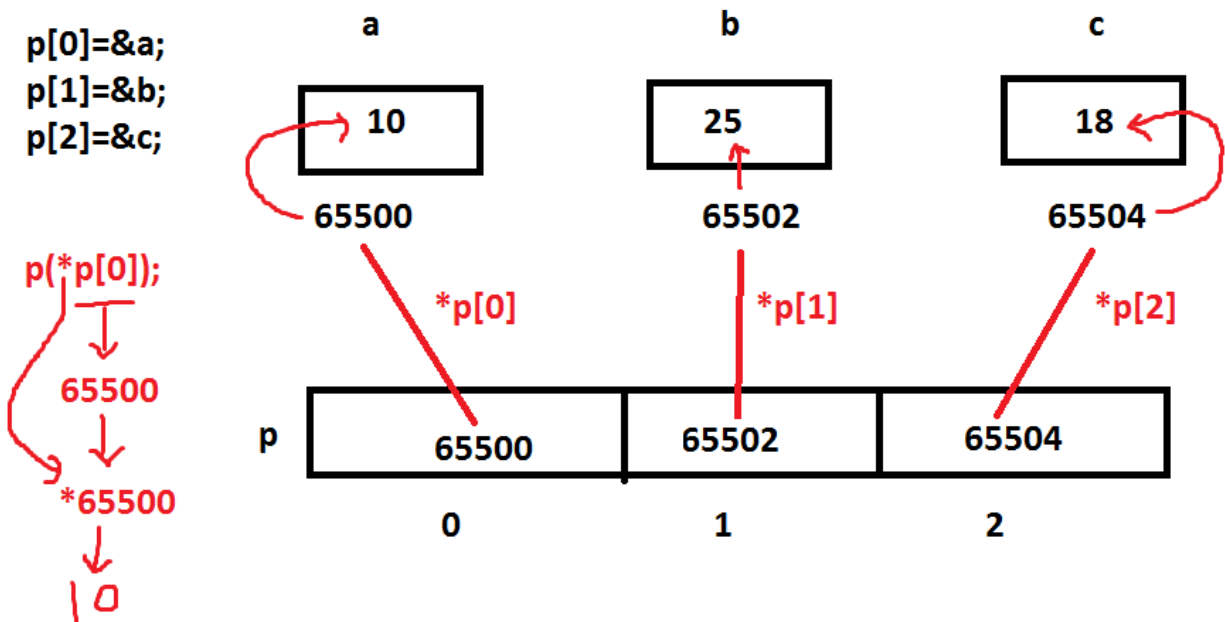
```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 9 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10, b=25, c=18, *p[3],i;
clrscr();
p[0]=&a; p[1]=&b; p[2]=&c;
for(i=0;i<3;i++)printf("%c=%d\n",97+i, *p[i]);
getch();
}

Activate Windows
Go to PC settings to activate Windows.
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```

Windows taskbar icons: DEV, File Explorer, Calculator, Zm, Paint, VS Code, Chrome, Word, Excel. System clock: 05:05 PM, 11-Apr-25.


```
a=10
b=25
c=18
```

Activate Windows
Go to PC settings to activate Windows.



Pointer to array:

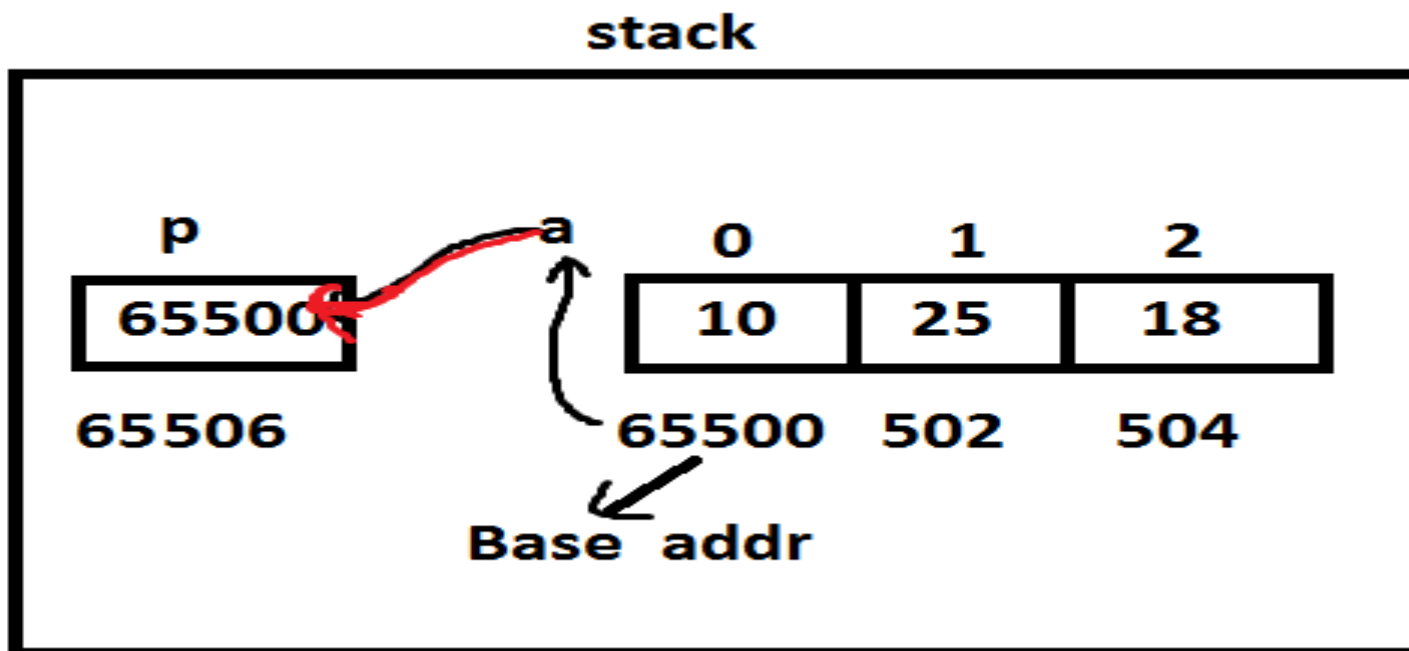
Array is implicit pointer. Due to this it holds the base cell addr [0 cell addr] implicitly. By assigning the array name or 0 cell addr to the pointer, we can handle array elements using the following syntax.

`*(ptrvariable + offset/index * sizeof(variable));`

Eg:

```
int a[3]={10, 25, 18}, *p, i;
```

```
p = a ; or p = &a[0]; or p = &a;
```



```
for(i=0;i<3;i++)
```

```
printf("%4d", *(p+i));
```

Here `*(p+i)` meaning is:

p is 65500

1. `*(p+0*2) → *65500 → value at 65500 → 10`
2. `*(p+1*2) → *65502 → value at 65502 → 25`

3. $*(p+2*2) \rightarrow *65504 \rightarrow \text{value at } 65504 \rightarrow 18$

Note: Here 2 is int size.

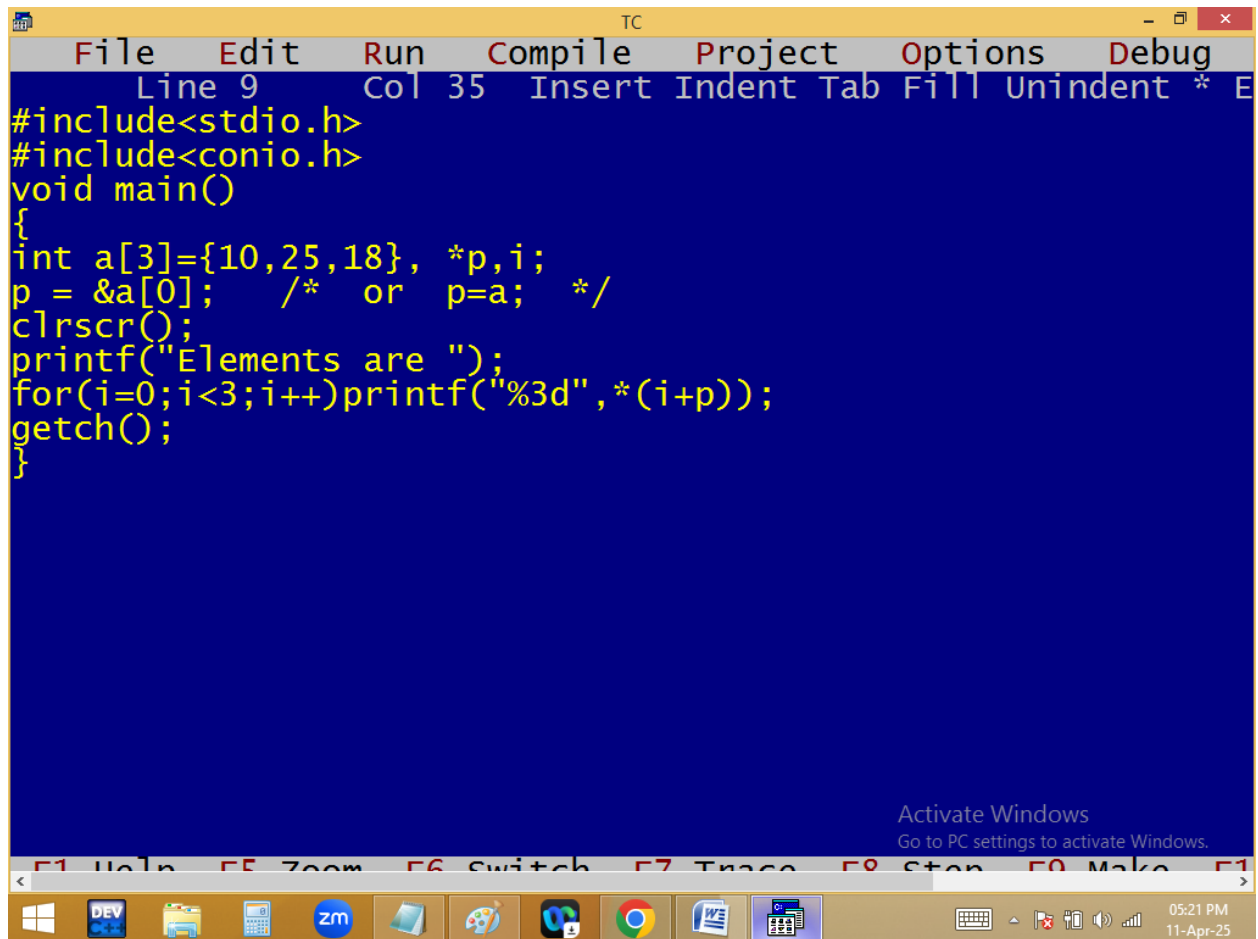
Eg:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int *p, a[3]={10,25,18}, i;
    clrscr();
    p = a; /* p=&a[0]; or p=&a; */
    printf("Elements are: ");
    for(i=0;i<3;i++)
        printf("%4d",*(p+i));
    getch();
}
```

Output: Elements are: 10 25 18

Note: We can access array elements using array / pointer in following ways.

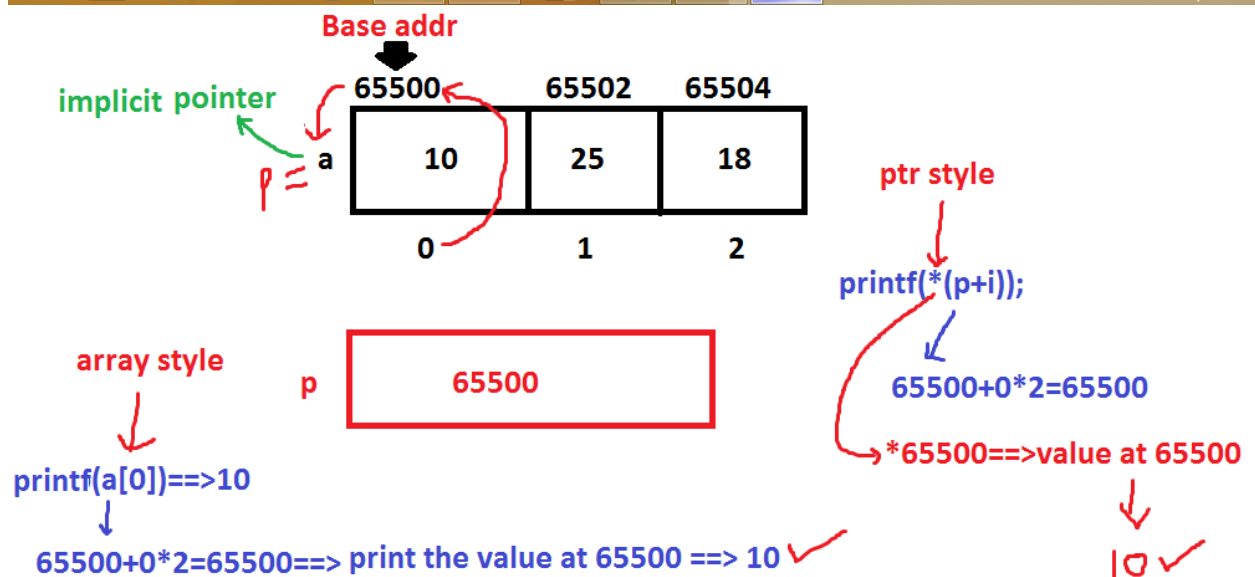
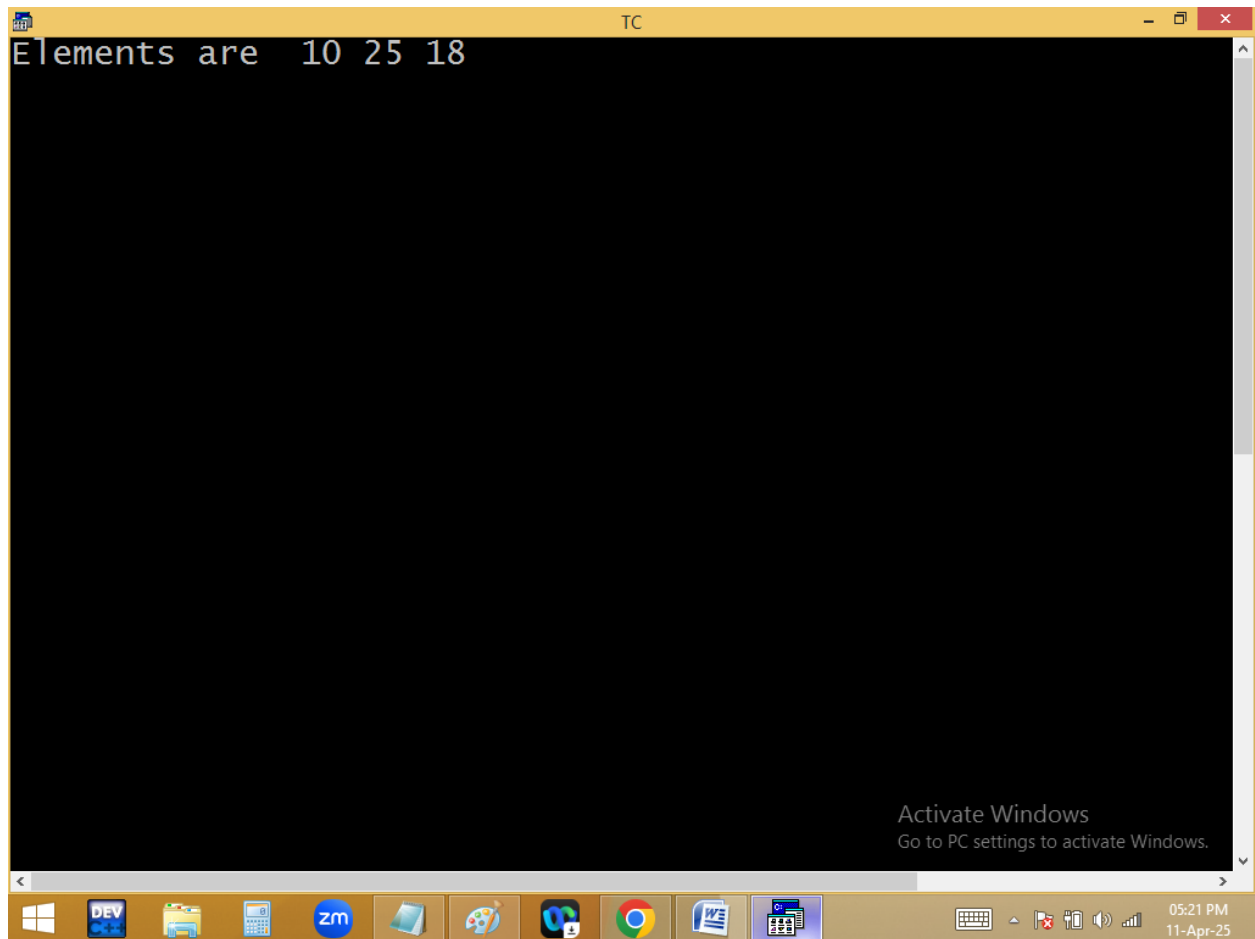
$a[i]$ / $i[a]$ / $p[i]$ / $i[p]$ / $*(p+i)$ / $*(a+i)$ / $*(i+p)$ / $*(i+a)$



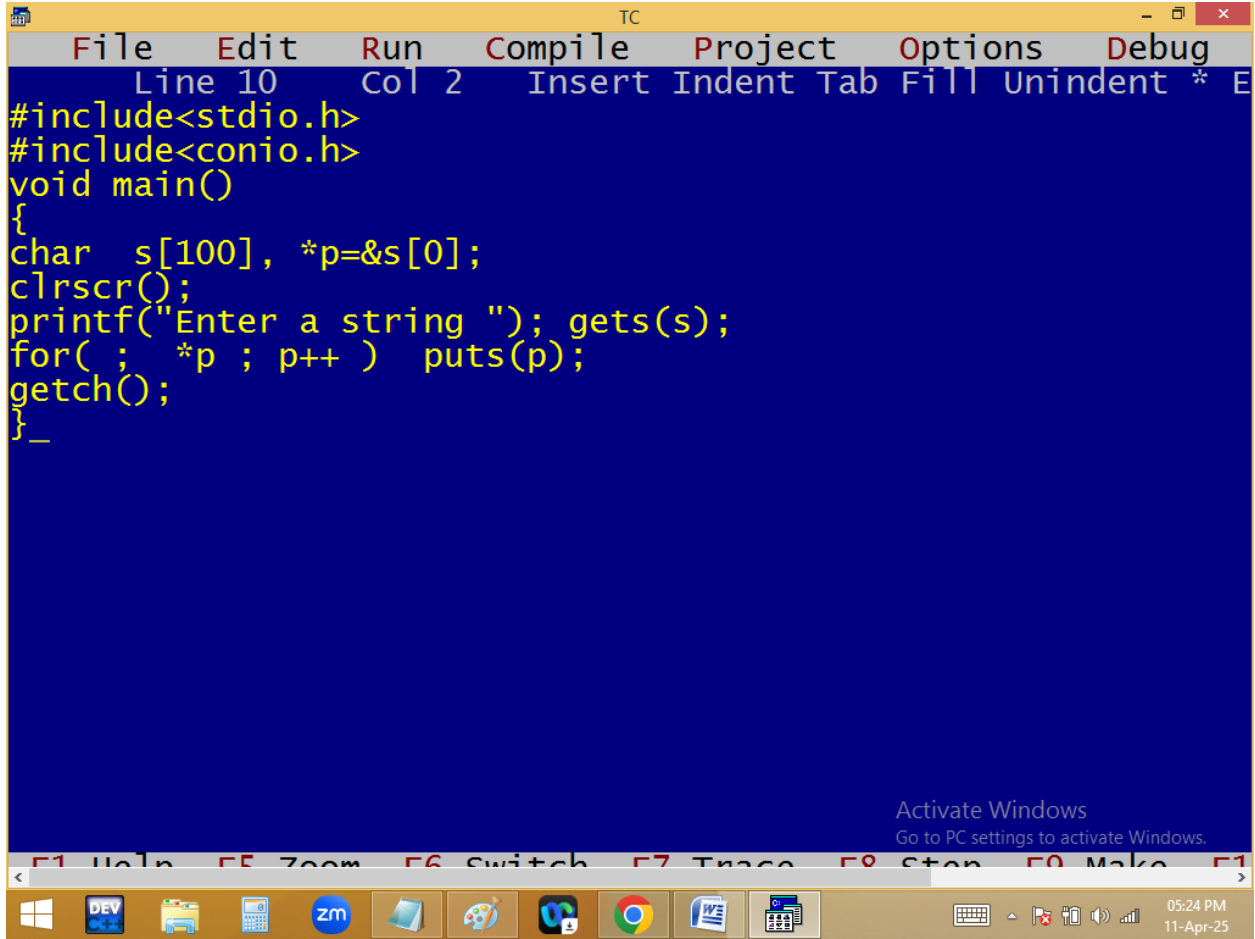
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 9 Col 35". The code editor contains the following C program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a[3]={10,25,18}, *p,i;
    p = &a[0]; /* or p=a; */
    clrscr();
    printf("Elements are ");
    for(i=0;i<3;i++)printf("%3d",*(i+p));
    getch();
}
```

At the bottom of the window, there is a taskbar with various application icons including Windows, DEV, File Explorer, Calculator, Zoom, and others. The system tray on the right shows the time as 05:21 PM and the date as 11-Apr-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

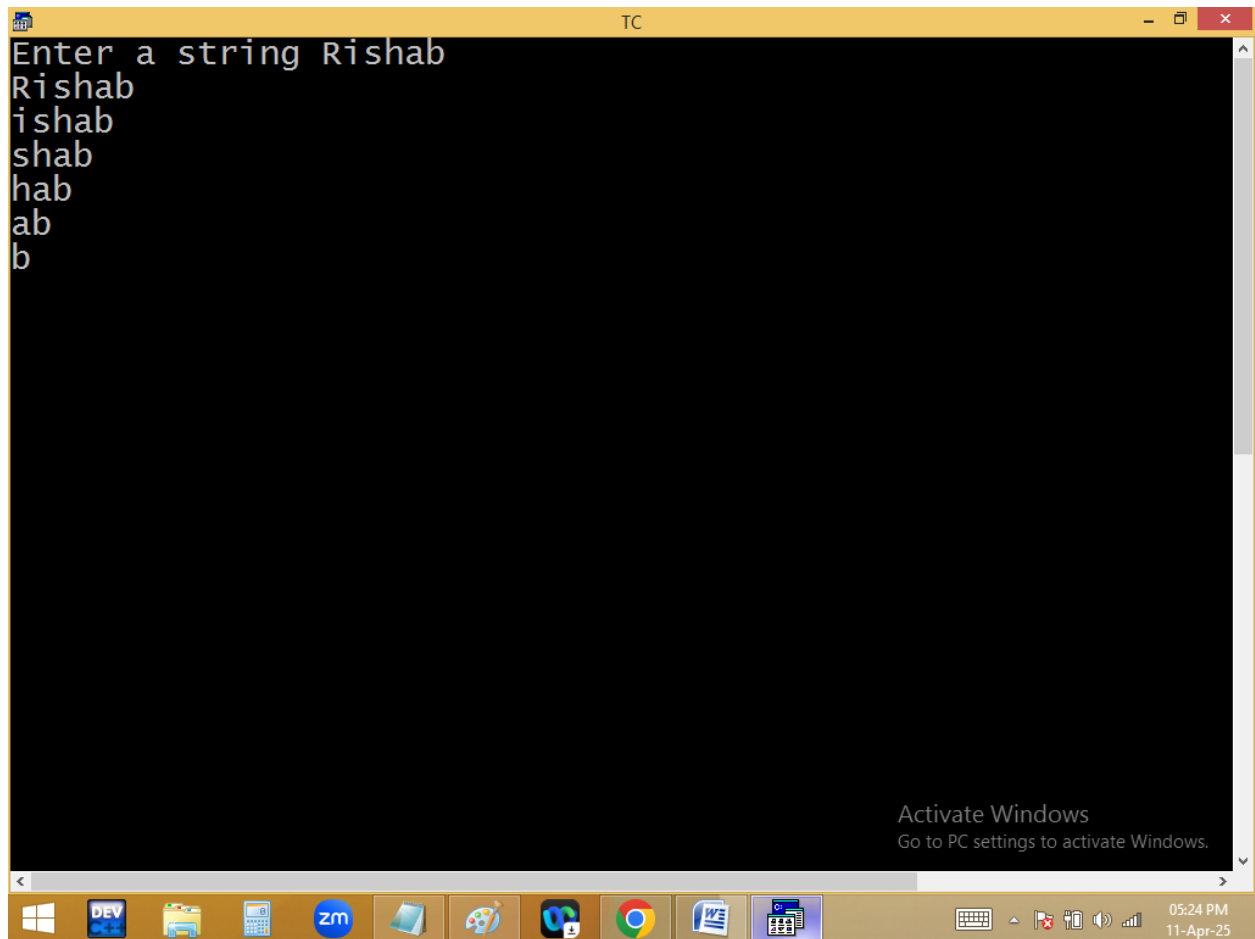


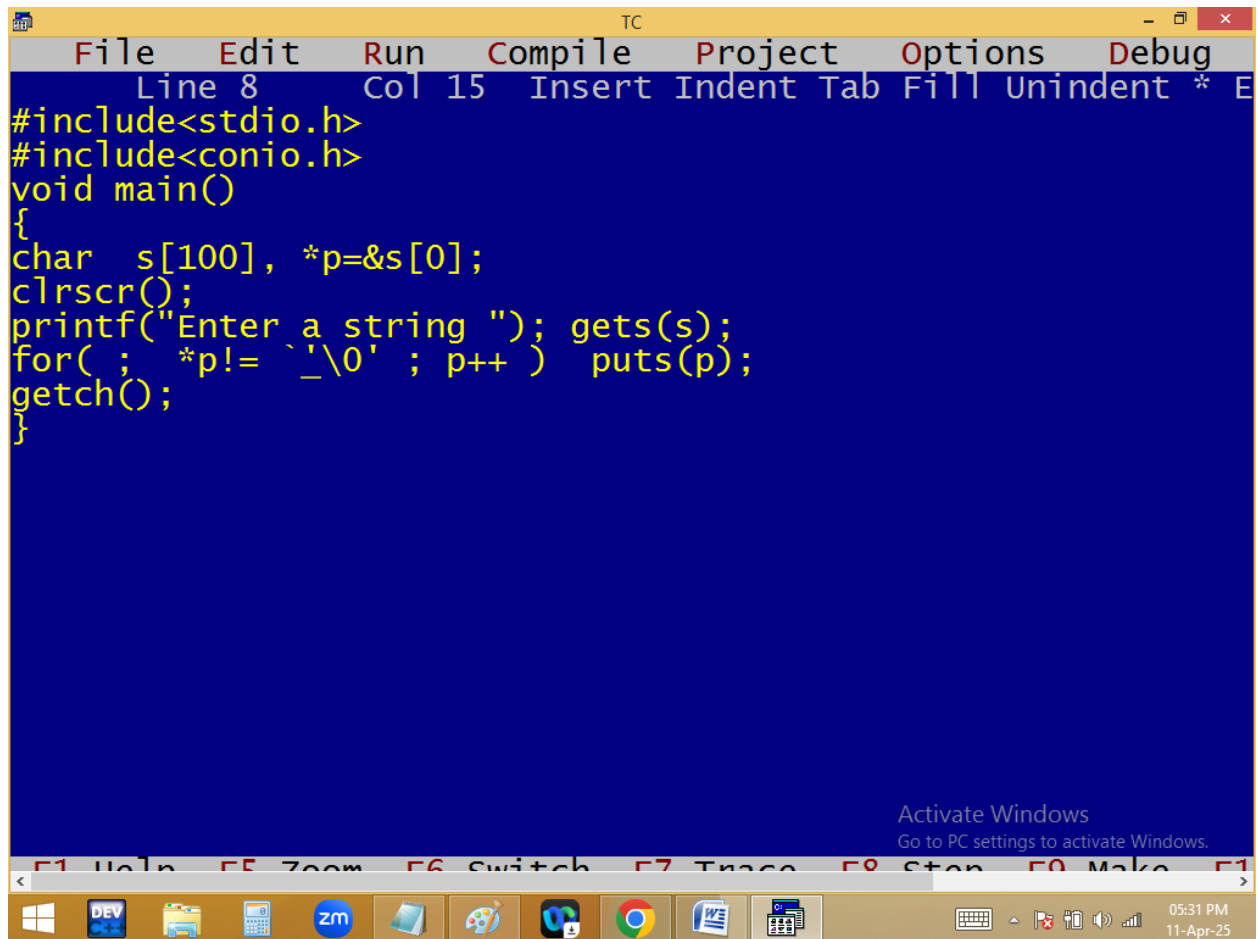
Pointer to string:



```
#include<stdio.h>
#include<conio.h>
void main()
{
char  s[100], *p=&s[0];
clrscr();
printf("Enter a string "); gets(s);
for( ; *p ; p++ ) puts(p);
getch();
}_
```

Activate Windows
Go to PC settings to activate Windows.





The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 8" and "Col 15". The main editing area has a blue background and contains the following C code:

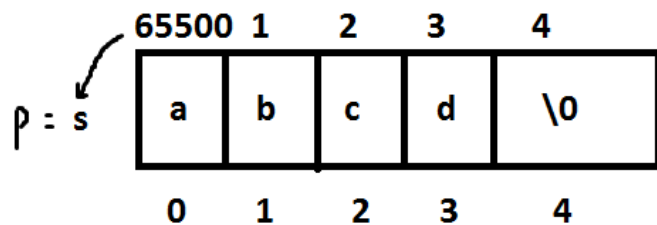
```
#include<stdio.h>
#include<conio.h>
void main()
{
char  s[100], *p=&s[0];
clrscr();
printf("Enter a string "); gets(s);
for( ; *p!= '\0' ; p++ ) puts(p);
getch();
}
```

At the bottom of the window, there is a taskbar with various application icons including Windows, DEV, File Explorer, Calculator, Zoom, and others. The system tray on the right shows the time as 05:31 PM and the date as 11-Apr-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.


```
Enter a string sindhu
sindhu
indhu
ndhu
dhu
hu
u
_
```

Activate Windows
Go to PC settings to activate Windows.

for(i=0; s[i];i++);



***p!='\0'**

for(; *p ; p++) puts (p);



65500 to \0 ==> abcd

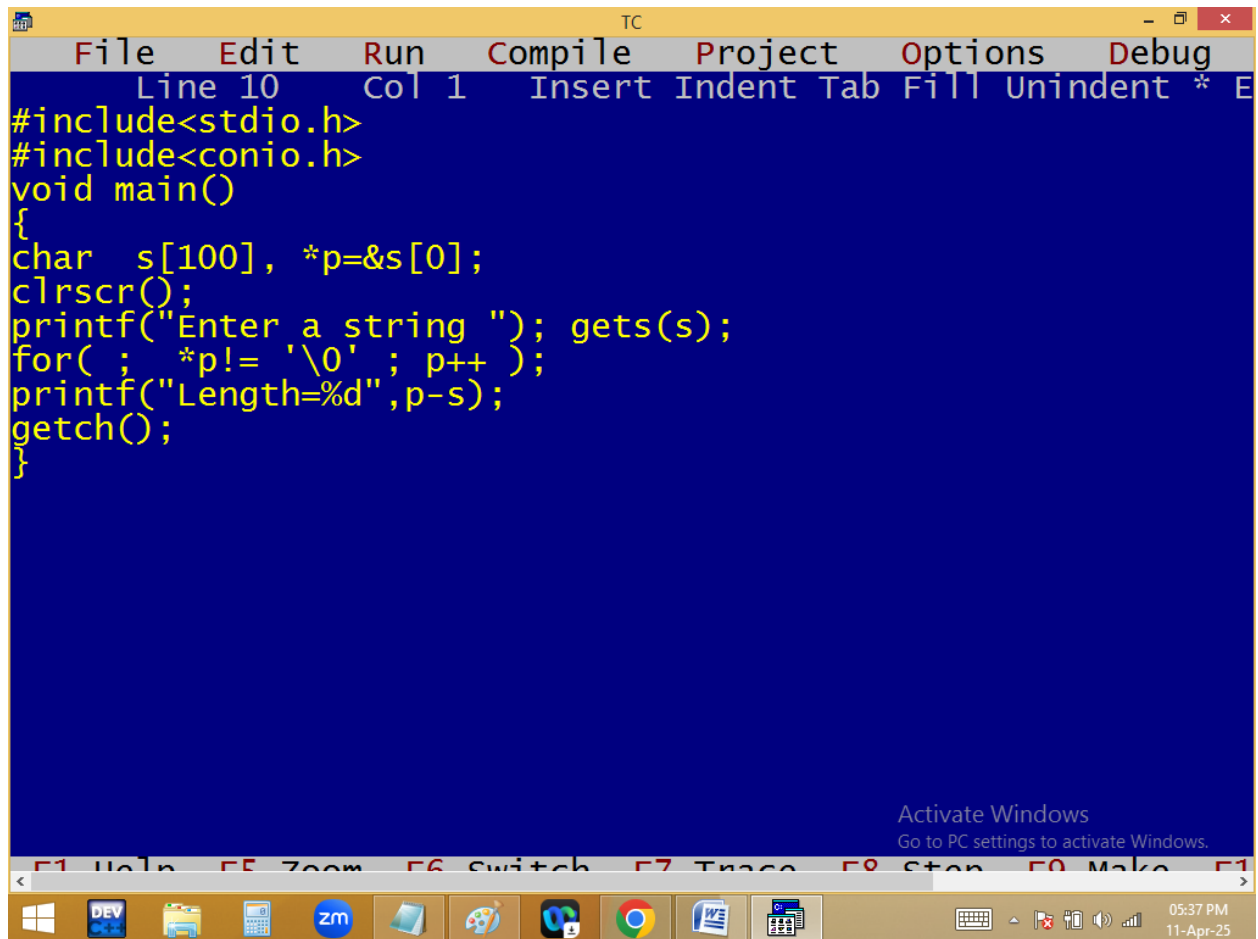
65501 to \0 ==> bcd

65502 to \0 ==> cd

65503 to \0 ==> d

65504 = \0 != \0 <== **False**

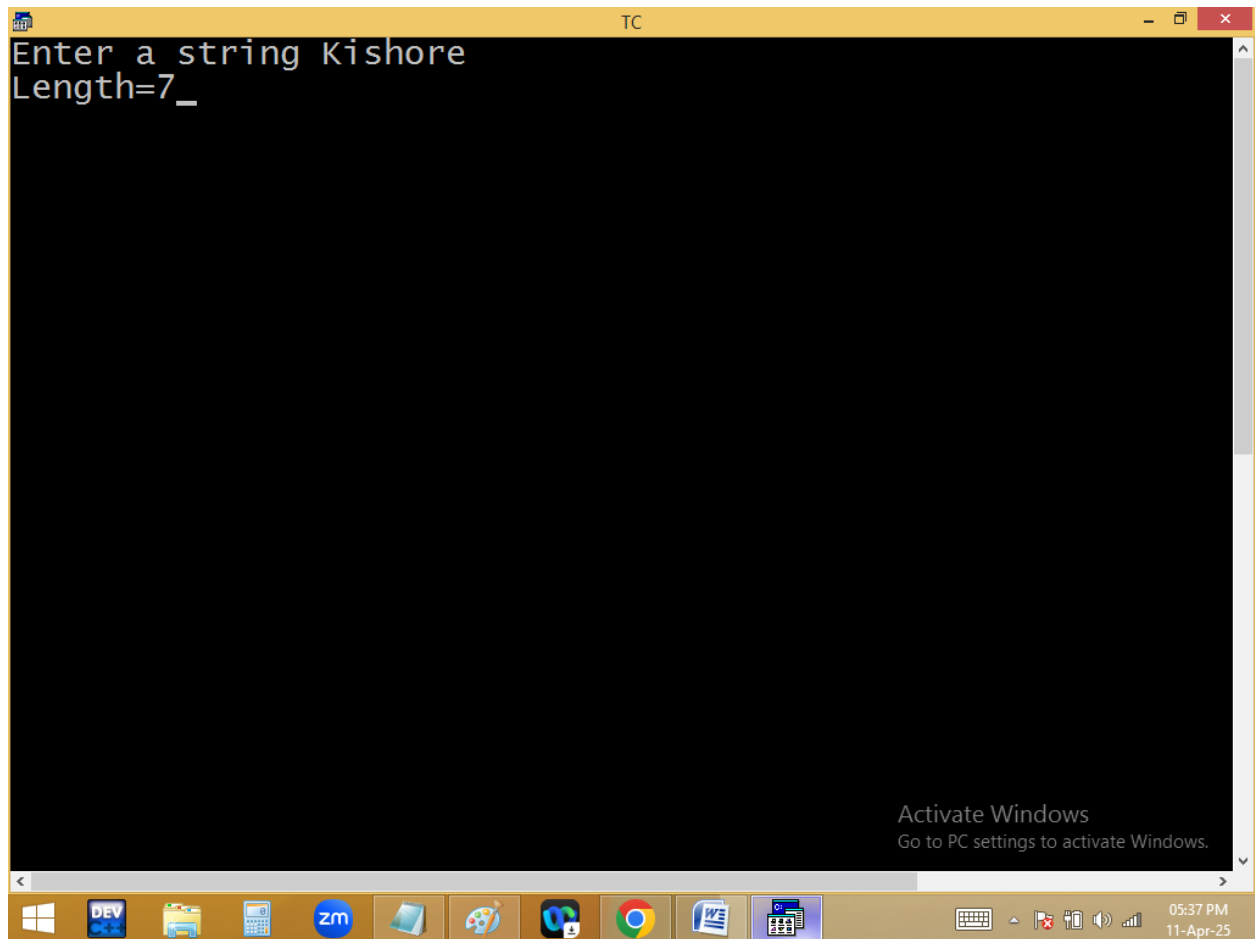
Find string length using pointer only?



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 10 Col 1" and lists keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The main editing area has a blue background and contains the following C code:

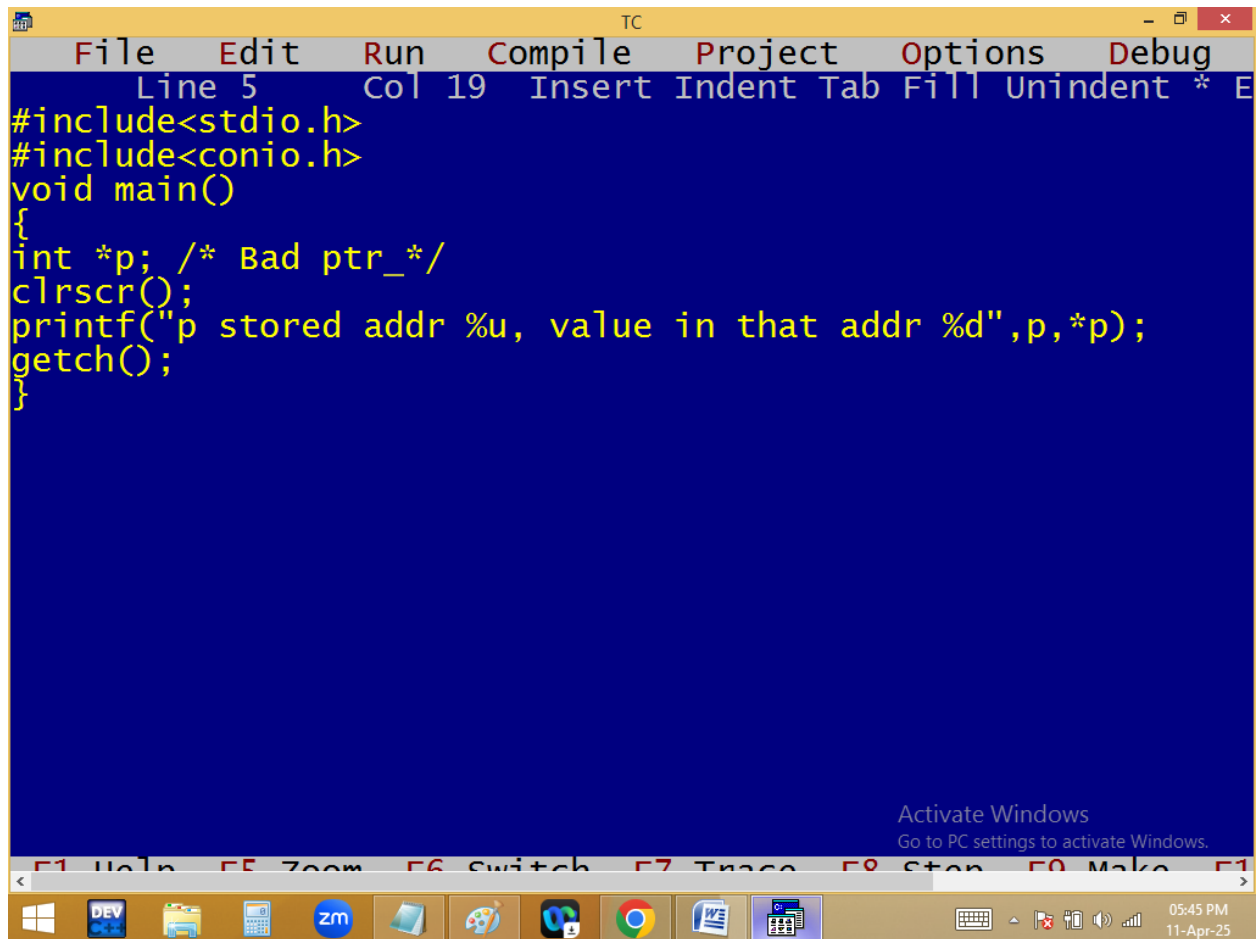
```
#include<stdio.h>
#include<conio.h>
void main()
{
char  s[100], *p=&s[0];
clrscr();
printf("Enter a string "); gets(s);
for( ; *p!= '\0' ; p++ );
printf("Length=%d",p-s);
getch();
}
```

At the bottom of the window, there is a taskbar with various application icons including Windows, DEV, File Explorer, Calculator, Zoom, and others. The system tray on the right shows the time as 05:37 PM and the date as 11-Apr-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.



Bad / wild pointer?

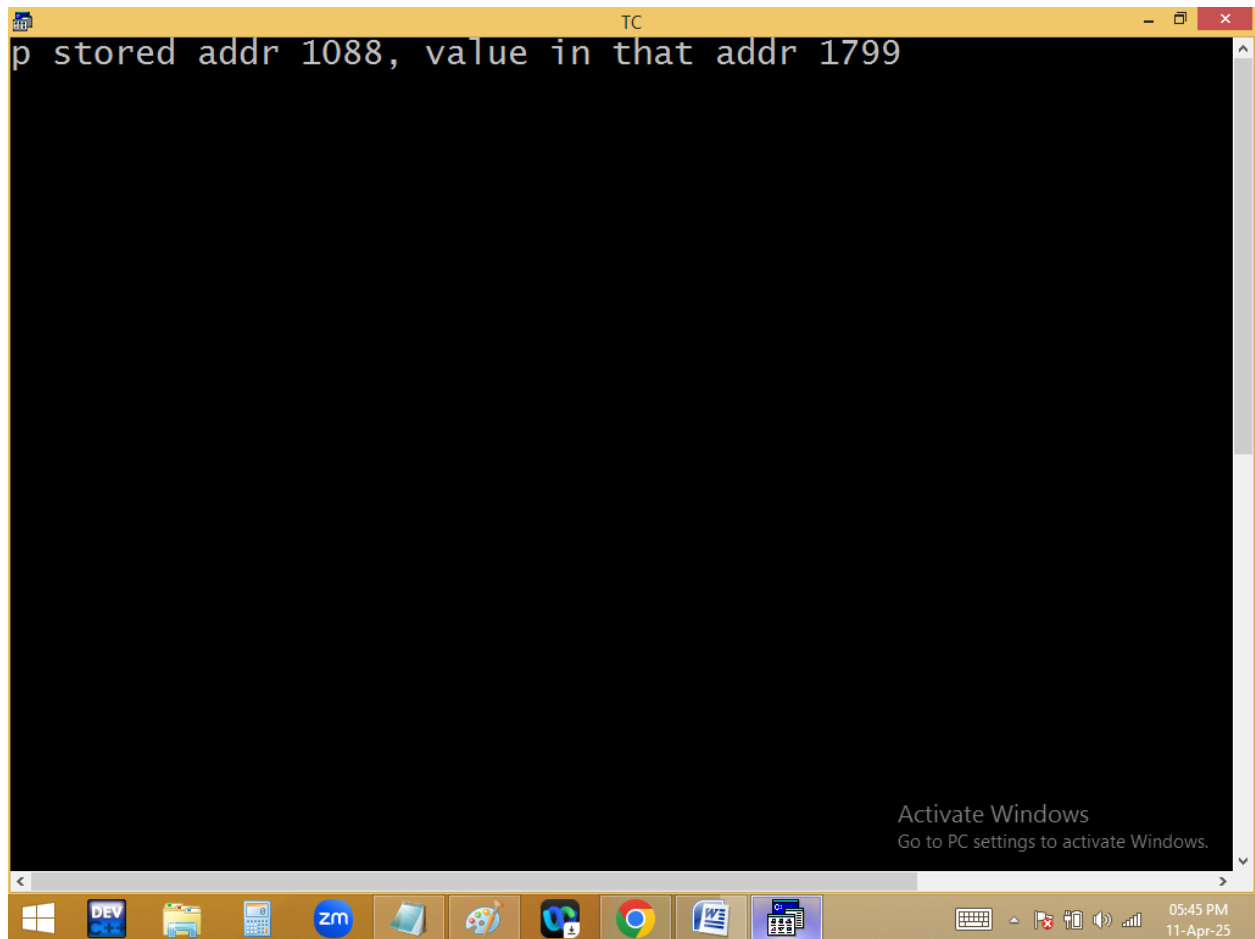
A pointer is declared but not initialized. In this situation the pointer takes some unknown address and value. This kind of pointer is called bad or wild pointer.



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 5 Col 19" and lists keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The main editing area has a blue background and contains the following C code:

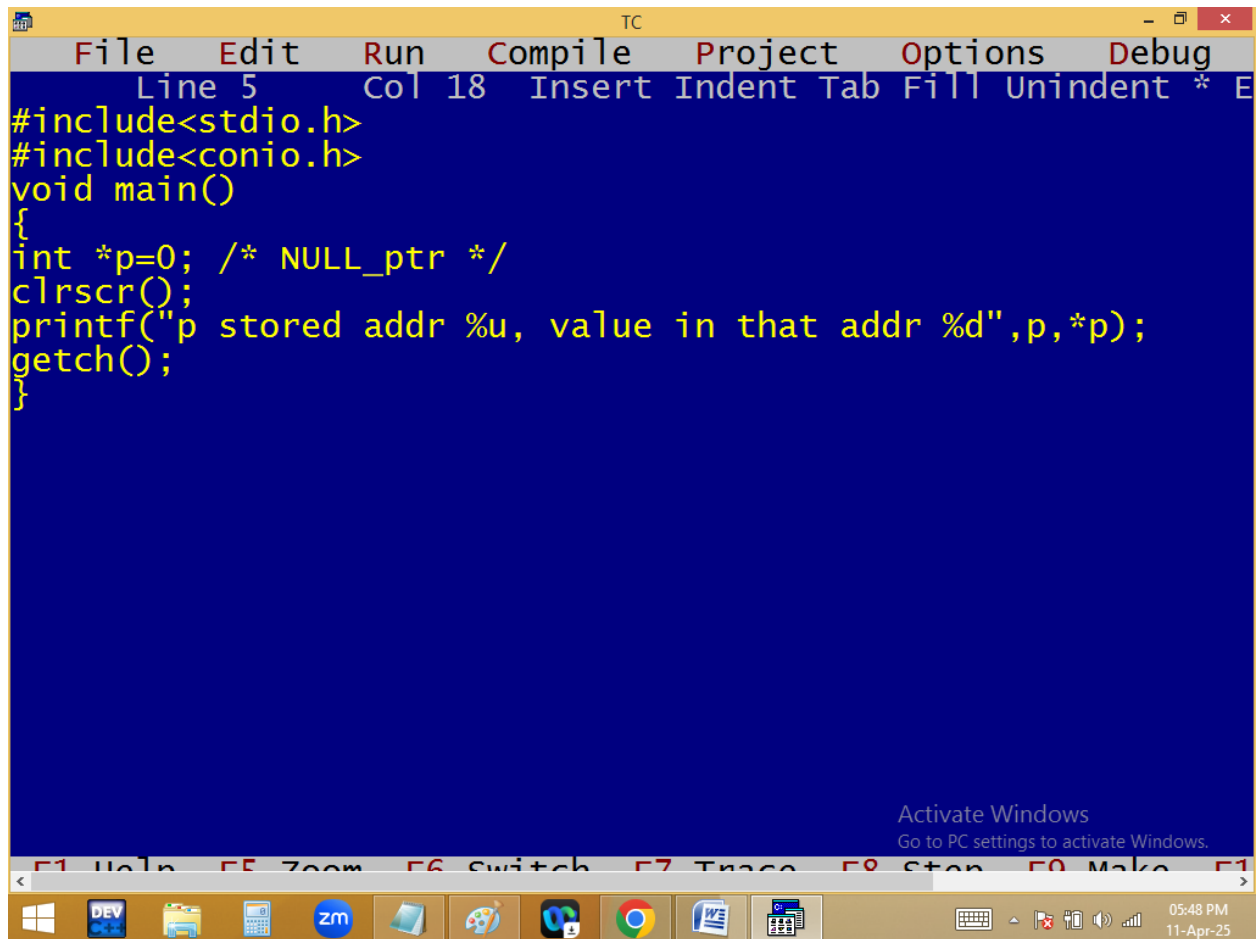
```
#include<stdio.h>
#include<conio.h>
void main()
{
int *p; /* Bad ptr_*/
clrscr();
printf("p stored addr %u, value in that addr %d",p,*p);
getch();
}
```

At the bottom of the IDE window, there is a message: "Activate Windows Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, DEV, File Explorer, Calculator, Zoom, a folder icon, Paint, VS Code, Google Chrome, Word, and Excel. The system tray on the right shows the date and time: "05:45 PM 11-Apr-25".



The image shows a screenshot of a Turbo C++ (TC) compiler window. The window has a yellow title bar with the text "TC" in the center. The main area is black with white text that reads "p stored addr 1088, value in that addr 1799". In the bottom right corner of the window, there is a message that says "Activate Windows" and "Go to PC settings to activate Windows." Below the window is a Windows taskbar with various icons including the Start button, DEV, File Explorer, Calculator, Zoom, and several application icons. The system clock in the bottom right corner shows "05:45 PM" and "11-Apr-25".

NULL pointer: When a pointer is initialized with NULL or 0 it is called NULL pointer. It is used to avoid the bad or dangling pointers.

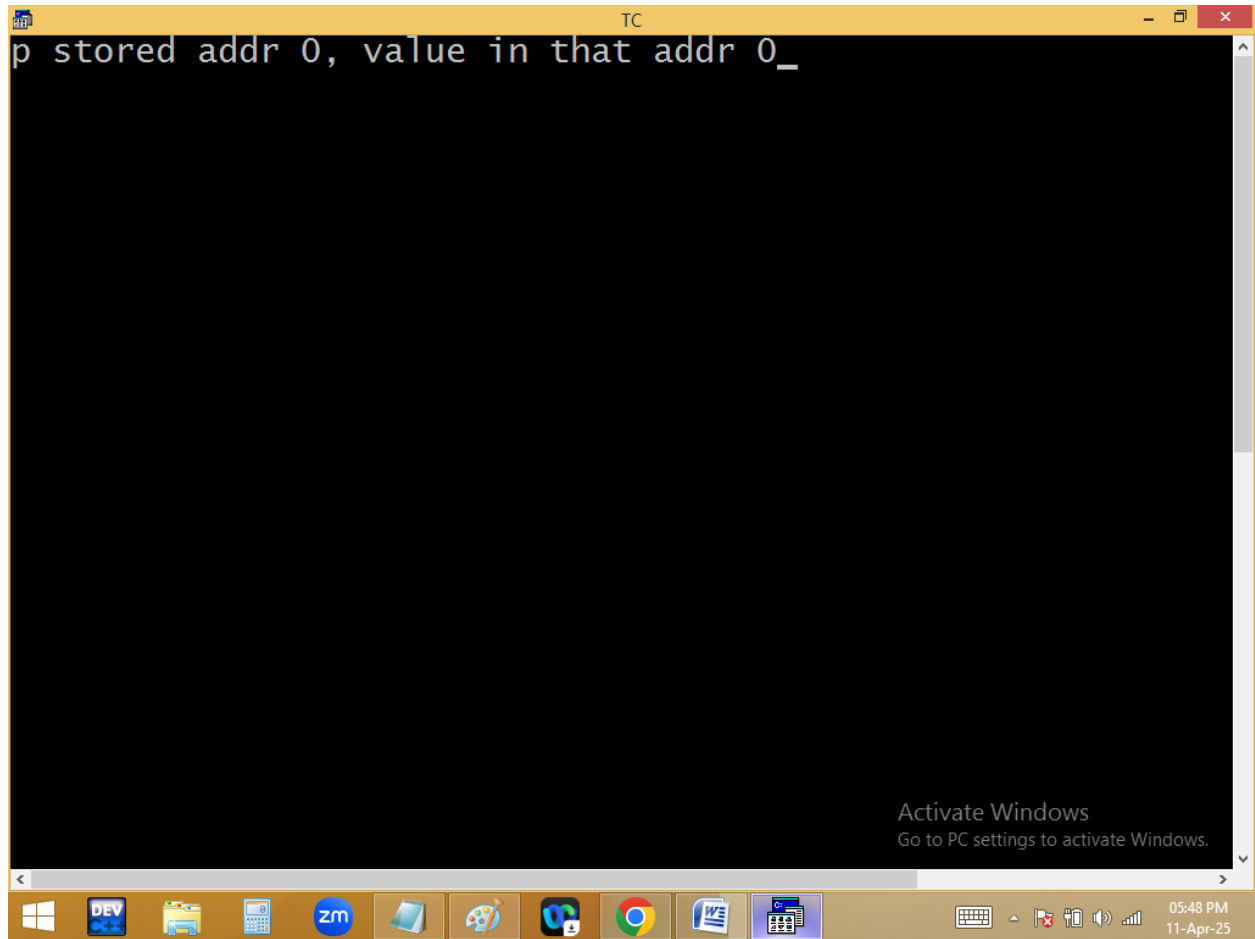


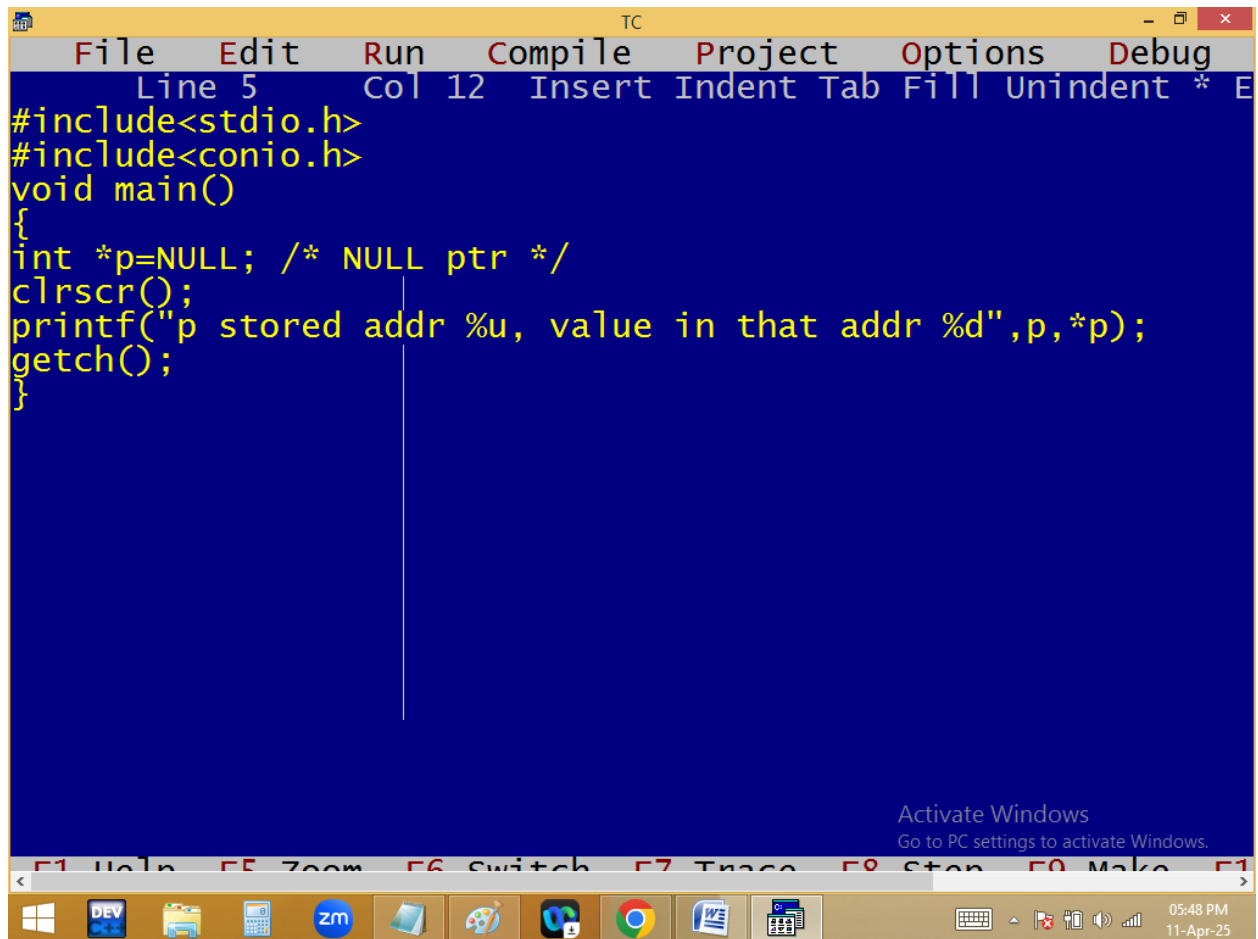
```
TC
File Edit Run Compile Project Options Debug
Line 5 Col 18 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int *p=0; /* NULL_ptr */
clrscr();
printf("p stored addr %u, value in that addr %d",p,*p);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

Windows taskbar: DEV, File Explorer, Calculator, ZOOM, Paint, VS Code, Chrome, Word, Excel, System tray (05:48 PM, 11-Apr-25)





The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 5", "Col 12", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and "* E". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int *p=NULL; /* NULL ptr */
clrscr();
printf("p stored addr %u, value in that addr %d",p,*p);
getch();
}
```

At the bottom of the IDE window, there is a message: "Activate Windows. Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, DEV, File Explorer, Calculator, Zoom, a folder icon, Paint, VS Code, Google Chrome, Word, and Excel. The system tray on the right shows the date and time: "05:48 PM 11-Apr-25".

